$$\widetilde{\Omega}^2(t) = -\frac{2}{M}\int_0^t \mathrm{d}\tau\, \eta(\tau)\cos\left(\Omega\tau\right), \tag{5.42a}$$

$$\gamma(t) = \frac{1}{M\Omega}\int_0^t \mathrm{d}\tau\, \eta(\tau)\sin\left(\Omega\tau\right), \tag{5.42b}$$

$$D(t) = \int_0^t \mathrm{d}\tau\, \nu(\tau)\cos\left(\Omega\tau\right), \tag{5.42c}$$

$$f(t) = -\frac{1}{M\Omega}\int_0^t \mathrm{d}\tau\, \nu(\tau)\sin\left(\Omega\tau\right). \tag{5.42d}$$